

SFT V11.11

The Standard Model Interface Ledger

Mass, Spin, Stress-Energy, and Thermal History
as Boundary-Maintenance Inputs

**SM dynamics supply the histories.
SFT accounts for their boundary maintenance.**

Working Draft v0.2

May 2026

This document is a layer-discipline note. It defines how the Standard Model can source Solvency Field Theory without being replaced by it. It is intentionally conservative: it states what SFT reads from the Standard Model, what SFT adds as boundary-maintenance accounting, and what this note does not claim to derive.

Contents

Abstract	2
1 The Interface Plate: SFT Plugged into the Standard Model	2
2 The Layer Discipline	2
3 The Master Interface Equation	3
4 Higgs Contact and the Rest-Clock Ledger	3
5 One Tick, Two Faces	4
6 Stress-Energy as Operation Density	5
7 Settlement Lag: Demand Is Not Settlement	5
8 Spin as a Holonomy Interface	6
9 Thermal Species Inventory and Solvency Features	7
10 The Interface Table	8
11 What This Note Establishes	8
12 What This Note Does Not Establish	8
13 Why This Matters for the V11 Stack	9
14 Conclusion	9
A Source Basis Within the V11 Archive	10

Abstract

The Standard Model is the most successful local field-and-interaction engine in physics. It specifies the known particle content, gauge interactions, Higgs mechanism, fermion masses through Yukawa contact, spin representations, and thermal species inventory. Solvency Field Theory does not replace that engine. Instead, V11.11 defines a ledger interface above it.

The central claim of this note is that the Standard Model supplies the inventory of histories, while SFT supplies the boundary-maintenance accounting required to keep those histories solvent. Mass becomes a maintained Compton rest-clock burden. Stress-energy becomes operation density and boundary write demand. Spin becomes a holonomy closure class of the internal lag-field frame. Finite-temperature species inventories become solvency loads and transition features. Settlement lag converts changing microscopic demand into scalar and tensor residuals.

In this form, SFT is not a competitor to the Standard Model Lagrangian. It is an accounting layer sourced by Standard Model outputs. The Standard Model says what exists and how it interacts. SFT asks what it costs the boundary to keep those histories written.

1 The Interface Plate: SFT Plugged into the Standard Model

The compact version is

$$\boxed{\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SM}} + \mathcal{L}_{\text{SFT}}[\text{SM outputs}]} \quad (1)$$

Spelled out as a socket map,

$$\boxed{\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SM}} + \mathcal{L}_{\text{SFT}}[T_{\text{SM}}^{\mu\nu}, \{m_i\}, \{y_i\}, \{s_i\}, \{g_i\}, v(T), \{\rho_i(T), p_i(T)\}, \Lambda, C_{\min}]} \quad (2)$$

The deliberately explicit Standard Model side is

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} - \frac{1}{4}W_{\mu\nu}^I W^{I\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} \quad (3)$$

$$+ \sum_{\psi} \bar{\psi} i \gamma^\mu D_\mu \psi + (D_\mu H)^\dagger (D^\mu H) - V(H) \quad (4)$$

$$- \left[\bar{Q}_L Y_u \tilde{H} u_R + \bar{Q}_L Y_d H d_R + \bar{L}_L Y_e H e_R + \text{h.c.} \right]. \quad (5)$$

The SFT layer then reads the Standard Model outputs through a maintenance functional:

$$\mathcal{L}_{\text{SFT}} = -\eta \nabla_\mu (\bar{\chi} + \psi) J_{\text{ops}}^\mu (T_{\text{SM}}^{\mu\nu}) + \Lambda_C C_{\min} - V_0 (e^{-\psi} - 1)^2 \quad (6)$$

$$+ \mathcal{L}_{\text{lag}}[\Lambda, \nabla \Lambda, \nabla \nabla \Lambda] + \mathcal{L}_{\text{hol}}[s_i, A_\theta] + \mathcal{L}_{\text{thermal}}[v(T), \rho_i(T), p_i(T)]. \quad (7)$$

This plate is the whole document in mathematical form. The Standard Model field engine remains intact. SFT is appended as a ledger functional of what the Standard Model already supplies: stress-energy, masses, Yukawa-generated rest clocks, spin representations, gauge-sector energy, and thermal species inventories.

The rest of this note explains each socket in the plate.

2 The Layer Discipline

The first rule of this interface is simple: do not mix the layers.

The Standard Model is the local quantum field theory. It carries the gauge groups, fields, couplings, scattering structure, fermion representations, Higgs mechanism, and renormalized observables. SFT does not attempt to rewrite that machinery here.

SFT instead asks a different question. Once the Standard Model has supplied a particle, mass, spin state, field configuration, stress-energy component, or thermal species inventory, what is the boundary cost of maintaining that history?

This is the same separation that appears throughout the V11 stack. A particle is not treated as a little object sitting in space. It is treated as a maintained history. A massive particle at rest is not idle. It is ticking internally at its Compton rate. A stress-energy distribution is not merely a source in a geometric equation. It is an operation-rate burden. A spinor is not a tiny rotating sphere. It is a phase-frame object with a nontrivial closure topology.

The Standard Model is the field-and-interaction engine. SFT is the boundary-maintenance ledger for the histories that engine produces.

This framing is intentionally conservative. V11.11 does not claim to derive the Standard Model from SFT. It defines the socket by which Standard Model outputs become SFT inputs.

3 The Master Interface Equation

A careless version of this program would try to insert SFT into the Standard Model Lagrangian and replace its internal terms. That is not the V11.11 claim.

The guarded action-level structure is

$$S_{\text{total}} = S_{\text{SM}} + S_{\text{SFT}}[g_{\mu\nu}, \psi, \bar{\chi}, J_{\text{ops}}^{\mu}(T_{\text{SM}}^{\mu\nu}), \Lambda, C_{\text{min}}, \dots]. \quad (8)$$

In interface-ledger form,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SM}} + \mathcal{L}_{\text{SFT}}[T_{\text{SM}}^{\mu\nu}, m_i, y_i, s_i, g_i, v(T), \rho_i(T), p_i(T)]. \quad (9)$$

The Standard Model side supplies

$$\{T_{\text{SM}}^{\mu\nu}, m_i, y_i, s_i, g_i, v(T), \rho_i(T), p_i(T)\}. \quad (10)$$

The SFT side translates those supplied quantities into

$$\{\nu_{C,i}, \rho_{\text{ops}}, \dot{a}_{\text{demand}}, \Lambda, \sigma_{ij}^{\text{lag}}, \mathcal{H}_s, \mathcal{U}(T)\}. \quad (11)$$

This is the interface ledger. It is not yet a completed unification. It is a disciplined map of responsibilities.

4 Higgs Contact and the Rest-Clock Ledger

The simplest place to see the interface is the Higgs/Yukawa sector.

For a Higgs-coupled elementary fermion, the Standard Model gives

$$m_f = \frac{y_f v}{\sqrt{2}}. \quad (12)$$

The Yukawa coupling y_f and Higgs vacuum expectation value v determine the fermion rest mass. SFT does not alter that statement. It takes the resulting mass and asks what rate of internal maintenance it implies.

The SFT translation is the Compton rest-clock rate:

$$\nu_{C,f} = \frac{m_f c^2}{h}. \quad (13)$$

Substituting the Higgs/Yukawa expression gives the direct contact ledger:

$$\nu_{C,f} = \frac{y_f v c^2}{\sqrt{2} h}. \quad (14)$$

So the SFT read is not that the Higgs mechanism is wrong. It is that Higgs contact sets the rate of a maintained internal clock. The Standard Model gives the mass; SFT assigns the maintenance burden.

For electroweak bosons and the Higgs excitation, the Standard Model similarly supplies

$$m_W = \frac{gv}{2}, \quad m_Z = \frac{\sqrt{g^2 + g'^2} v}{2}, \quad m_H = \sqrt{2\lambda} v. \quad (15)$$

SFT then reads each rest mass through

$$\nu_{C,i} = \frac{m_i c^2}{h}. \quad (16)$$

This is why the Higgs sector is a natural SFT socket. When the Higgs field gives a particle a rest mass, it also gives the boundary a rest-clock obligation.

Guardrail. This section does not derive Yukawa values. It does not claim that all mass is Higgs mass. Hadronic masses are dominated by QCD confinement and binding. Photons and gluons do not carry Higgs rest clocks, although they still carry operation rate through energy. Neutrino masses require a separate treatment and should not be smuggled into the minimal Higgs-contact ledger without care.

5 One Tick, Two Faces

A Compton clock should not be interpreted as a little mechanical clock inside the particle. It is the phase rate associated with maintaining a rest-mass history.

In ordinary language, a massive particle at rest has no spatial velocity. In SFT language, that does not make it inactive. Its full budget is internal. The phase advances as

$$d\phi = \omega_C d\tau = \frac{mc^2}{\hbar} d\tau. \quad (17)$$

The V11 history-axis interpretation is

$$\text{one tick} = \text{one irreversible history increment}. \quad (18)$$

That history increment has a local face and a boundary face.

The local face is proper-time/phase advance:

$$dN_i \sim \nu_{C,i} d\tau. \quad (19)$$

The boundary face is capacity growth:

$$dA_i = \xi \ell_P^2 dN_i. \quad (20)$$

Thus a rest clock is not just a local oscillation. It is a maintained history whose ticks must be written. The particle sees proper time. The boundary sees write demand.

This is the first major conceptual bridge between Standard Model particle content and SFT maintenance accounting.

6 Stress-Energy as Operation Density

The Higgs-contact ledger handles rest clocks. The broader Standard Model stress-energy tensor handles everything else.

The Standard Model supplies

$$T_{\text{SM}}^{\mu\nu}. \quad (21)$$

For a perfect-fluid source in the SFT accounting language, the operation density is written

$$\rho_{\text{ops}} = \frac{\epsilon + 3p}{h}. \quad (22)$$

The factor $\epsilon + 3p$ is important. Radiation and relativistic pressure are not passive. They carry stronger operation burden than pressureless matter. In the covariant V11 action language, this is the local scalar contraction of stress-energy with the observer frame and spatial projector:

$$\rho_{\text{ops}} = \frac{T_{\mu\nu} u^\mu u^\nu + h_{\mu\nu} T^{\mu\nu}}{h_P}. \quad (23)$$

The local write-demand density is then

$$\dot{a}_{\text{demand}}(x, t) = \xi \ell_P^2 \rho_{\text{ops}}(x, t). \quad (24)$$

This is the second major socket:

$$T_{\text{SM}}^{\mu\nu} \longrightarrow \rho_{\text{ops}} \longrightarrow \dot{a}_{\text{demand}}. \quad (25)$$

The Standard Model says what the fields are doing. SFT counts how expensive those histories are to maintain.

7 Settlement Lag: Demand Is Not Settlement

Once operation demand exists, the boundary still has to settle it. SFT distinguishes demand from completed settlement.

The lag residual is

$$\Lambda(x, t) = A_{\text{demand}}(x, t) - A_{\text{settled}}(x, t). \quad (26)$$

The minimal scalar settlement equation is

$$\partial_t \Lambda = \dot{a}_{\text{demand}} - \frac{\Lambda}{\tau_{\text{lag}}}. \quad (27)$$

A causal-diffusive extension is

$$\partial_t \Lambda + \frac{\Lambda}{\tau_{\text{lag}}} - D_\Lambda \nabla^2 \Lambda = \dot{a}_{\text{demand}}. \quad (28)$$

This is the third bridge. Changing microscopic demand does not become macroscopic geometry instantly. There is a finite settlement channel. Where the channel is stressed, residuals appear.

A scalar force proxy can be written

$$f_{\text{scalar}} \sim -\nabla \Lambda. \quad (29)$$

The trace-free tensor structure is

$$\Lambda_{ij}^{\text{TF}} = \left(\partial_i \partial_j - \frac{1}{3} \delta_{ij} \nabla^2 \right) \Lambda. \quad (30)$$

A candidate lag stress is then

$$\sigma_{ij}^{\text{lag}} \sim -\eta_\Lambda \Lambda_{ij}^{\text{TF}}. \quad (31)$$

This should be read carefully. It is not ordinary viscosity. It is not a completed microscopic transport derivation. It is a settlement residual generated when operation demand and boundary settlement are not identical in time.

In the V11.10 late-time bridge, this distinction mattered. The physical history was not a switch. It was a handoff. V11.11 explains why that kind of handoff is natural: microscopic source demand and boundary settlement are related, but not instantaneously identical.

8 Spin as a Holonomy Interface

Spin is often described badly in ordinary language. It is not a little ball spinning around an axis. In the Standard Model and relativistic quantum theory, spin is representation-theoretic.

The important group-theoretic fact is that spinors live in $SU(2)$, the double cover of $SO(3)$:

$$2\pi : \psi \rightarrow -\psi, \quad 4\pi : \psi \rightarrow \psi. \quad (32)$$

SFT translates this into lag-field holonomy language.

Mass sets the internal ticking rate:

$$m \rightarrow \nu_C. \quad (33)$$

Spin sets the closure topology:

$$s \rightarrow \text{phase-frame holonomy class}. \quad (34)$$

Let an internal phase-frame connection be

$$\frac{dU}{d\theta} = i A_\theta U. \quad (35)$$

The holonomy around the internal loop is

$$\mathcal{H} = \exp \left(i \oint A_\theta d\theta \right). \quad (36)$$

Integer closure corresponds to

$$\oint A_\theta d\theta = 2\pi n \quad \Rightarrow \quad \mathcal{H} = +1. \quad (37)$$

Spinor half-twist closure corresponds to

$$\oint A_\theta d\theta = (2n + 1)\pi \quad \Rightarrow \quad \mathcal{H} = -1. \quad (38)$$

This gives the fourth socket:

$$\text{SM spin representation} \longrightarrow \text{SFT holonomy closure class}. \quad (39)$$

The SFT claim is not that spin has been fully derived from scratch. The claim is that the lag-field representation gives a natural accounting language for the closure topology that spinors already require.

Guardrail. This does not derive spin-statistics. It does not derive Pauli exclusion. It does not compute the QED anomalous magnetic moment. At most, the V11.9 lineage shows that leading $g = 2$ behavior can be placed at the Pauli/Dirac interface level. The anomaly remains a higher-order quantum-field calculation.

9 Thermal Species Inventory and Solvency Features

At finite temperature, the Standard Model does not present a static particle ledger. The active species inventory changes with temperature.

At temperature T , the relevant inputs include

$$\rho_i(T), \quad p_i(T), \quad m_i(T), \quad g_i, \quad v(T). \quad (40)$$

When the Higgs vacuum expectation value turns on across the electroweak transition, rest-clock obligations reorganize. Particles that were effectively massless in the symmetric phase acquire rest-clock burden in the broken phase.

A schematic SFT thermal ledger can be written

$$\mathcal{U}(T) = \sum_i w_i(T) \mathcal{C}_i(T), \quad (41)$$

where $\mathcal{C}_i(T)$ is the species operation/rest-clock contribution and $w_i(T)$ is the SFT ledger weighting.

Candidate forcing channels include

$$F_{d\mathcal{U}}, \quad F_{d\ln\mathcal{U}}, \quad F_{\text{mass}}. \quad (42)$$

The electroweak transition is therefore not merely a particle-physics event in SFT language. It is a ledger reorganization. The active operation inventory changes, the rest-clock budget changes, and the boundary settlement channel sees a feature.

This is why the Standard Model interface matters for baryogenesis and pre-recombination cosmology. SFT should not invent species. It should read the Standard Model thermal inventory and translate it into solvency forcing.

Guardrail. SFT does not derive CP violation here. It does not derive baryon-number violation. It does not replace finite-temperature electroweak field theory. The SFT contribution is the derived solvency-forcing channel that reads the thermal inventory supplied by the Standard Model.

10 The Interface Table

Standard Model source	SFT ledger translation	Resulting SFT object
Higgs/Yukawa mass, $m_f = y_f v / \sqrt{2}$	Rest-clock maintenance	$\nu_C = mc^2 / h$
Electroweak boson/Higgs masses	Heavy-sector clock burden	$\nu_{C,W}, \nu_{C,Z}, \nu_{C,H}$
Gauge radiation energy	Operation rate without rest clock	E/h contribution
Stress-energy, $T_{\text{SM}}^{\mu\nu}$	Operation density	ρ_{ops}
Operation density	Boundary write demand	$\dot{a}_{\text{demand}} = \xi \ell_P^2 \rho_{\text{ops}}$
Demand minus settlement	Lag residual	$\Lambda = A_{\text{demand}} - A_{\text{settled}}$
Gradients of lag residual	Scalar response	$-\nabla \Lambda$
Trace-free second derivatives	Tensor lag stress	σ_{ij}^{lag}
Spin representation	Closure topology	$\mathcal{H} = \exp(i \oint A_\theta d\theta)$
Finite-temperature species inventory	Solvency load	$\mathcal{U}(T) = \sum_i w_i \mathcal{C}_i$

11 What This Note Establishes

V11.11 establishes a layer-clean interface between the Standard Model and SFT.

It establishes that the Higgs/Yukawa sector can be read as a rest-clock ledger without altering the Standard Model mass mechanism.

It establishes that Standard Model stress-energy can be read as operation density and boundary write demand.

It establishes that finite settlement lag supplies a natural bridge between microscopic operation demand and macroscopic scalar/tensor residuals.

It establishes that spin can be represented in SFT as holonomy closure of an internal lag-field frame, while preserving the group-theoretic facts that standard quantum theory already requires.

It establishes that the finite-temperature Standard Model inventory supplies the correct input language for SFT solvency features.

Most importantly, it establishes that SFT should not be presented as replacing the Standard Model. SFT is a maintenance ledger sourced by the Standard Model.

12 What This Note Does Not Establish

1. A derivation of the Standard Model Lagrangian.
2. A derivation of gauge couplings.
3. A derivation of the number of generations.
4. A derivation of the Yukawa spectrum.
5. A solution to the neutrino mass problem.
6. A derivation of spin-statistics or Pauli exclusion.

7. A calculation of anomalous magnetic moments.
8. A full finite-temperature electroweak computation.
9. A final CMB likelihood result.

These are not defects of the note. They are layer boundaries. A useful interface ledger must say what it is not doing as clearly as it says what it is doing.

13 Why This Matters for the V11 Stack

V11.10 showed that late-time source histories, DESI/BAO residual direction, CLASS sound-horizon reconstruction, and the Hubble bridge can be placed into a clean reproducible chain. But the source histories in that chain were not magic. They were histories of matter, black holes, area response, and settlement channels.

V11.11 gives the deeper language for that move. Matter and radiation do not enter SFT as arbitrary phenomenological knobs. They enter as Standard Model and effective-field-theory source inventories. Their masses, pressures, phase histories, and thermal transitions determine the operation load. The SFT layer then asks how the boundary settles that load.

This is also why the pre-recombination CMB companion problem should be treated as a Standard Model interface problem, not merely a curve-fitting problem. The early-universe sound horizon depends on the thermal inventory, expansion history, and operation burden of relativistic and nonrelativistic species. If SFT modifies the pre-recombination expansion profile, the physically clean route is to derive that modification from the Standard Model thermal ledger and the SFT settlement law.

Thus V11.11 sits between V11.10 and the CMB companion paper. It explains what the sources are.

This is not only aspirational. Several V11-era computations already used this interface in practice. The V11.8 baryogenesis line treated the electroweak Standard Model inventory as a solvency-forcing source rather than as an invented SFT species list. The V12a-V12e pre-recombination sound-horizon reconstruction treated thermal history and expansion burden as a ledger problem whose target could be passed into CLASS. The V11.7 fluid-sector work treated matter/radiation source histories as operation-density inputs that generate settlement and lag responses. V11.11 names the common architecture underneath those runs: Standard Model and effective-field-theory inputs are read by the SFT maintenance ledger, producing quantitative boundary-settlement consequences.

14 Conclusion

The Standard Model says what exists and how it interacts. SFT asks what it costs the boundary to keep those histories solvent.

This is the simplest disciplined form of the Standard Model/SFT synthesis. Higgs contact becomes rest-clock burden. Stress-energy becomes operation density. Operation density becomes write demand. Demand minus settlement becomes lag. Spin becomes holonomy closure. Thermal species inventories become solvency features.

The result is not a replacement for particle physics. It is a ledger above particle physics.

In one line:

$$\boxed{\text{SM dynamics supply the histories; SFT accounts for their boundary maintenance.}} \quad (43)$$

A Source Basis Within the V11 Archive

The source basis for this synthesis is the V11 lineage rather than a new independent computational run. The main internal anchors are:

- **V11.3 V2 - Higgs Contact Ledger:** Higgs/Yukawa masses as Compton rest-clock burden, including guardrails on Yukawas, QCD mass, photons, gluons, and neutrinos.
- **V11.5 / V11.6 - One-Speed and Boundary-Time Duality:** rest-mass phase history, one tick as one irreversible history increment, and the local/boundary two-face interpretation.
- **V11.1B - Planck-Area Writes, Lag Stress, and Fluid Coefficients:** operation density, write demand, settlement residual, scalar lag gradients, and trace-free tensor lag stress.
- **V11.9 - Spin as Lag-Field Holonomy:** $SU(2)/SO(3)$ closure, spinor half-twist behavior, and the guarded Pauli/Dirac interface.
- **V11 Companion action notes:** repaired five-term SFT action and covariant operation-density architecture.
- **V11.8 baryogenesis lineage:** thermal electroweak inventory as a solvency-forcing source, with CP and baryon-number guardrails.
- **V11.10 late-time crossing and bridge work:** a frozen reproducible example of source histories processed through the SFT ledger into cosmological consequences.